

## Using Ratios

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- **Ratios are useful in assessing profitability, liquidity, and risk**
  - Highlight sources of competitive advantage and “red flag” potential trouble
- **Ratios must be compared to a benchmark**
  - Compare same firm across time (time-series analysis)
  - Compare firm to other firms or to industry (cross-sectional analysis)
- **Ratios are contextual**
  - Try to determine the underlying activity that the ratio represents to determine whether it is good or bad news
- **Key: Ratio analysis does not provide answers, but instead helps you ask better questions**

## Misusing ratios

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- **Standard ratios have multiple definitions**
  - There is no GAAP for ratio definitions
  - Use the same definition to make valid comparisons
- **Choosing the appropriate benchmark for comparison is important**
  - Major changes in the firm distort time-series analysis
  - Differences in business strategy, capital structure, or business segments distort cross-sectional analysis
  - Differences in accounting methods make all comparisons difficult
- **Ratios may be manipulated by managerial action**

# Return on Equity

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- Is a Net Income of \$10,000,000 good or bad?
  - Depends on level of investment required
- Return on Equity (ROE)
  - $ROE = \text{Net Income} / \text{Average Shareholders' Equity}$
  - The numerator represents how much return the company generated for shareholders during the year based on accrual accounting
  - The denominator represents the shareholders' investment in the company
    - Must take average of beginning and ending balances
  - Measures Return on Investment (ROI)
    - Should increase with the risk of the company

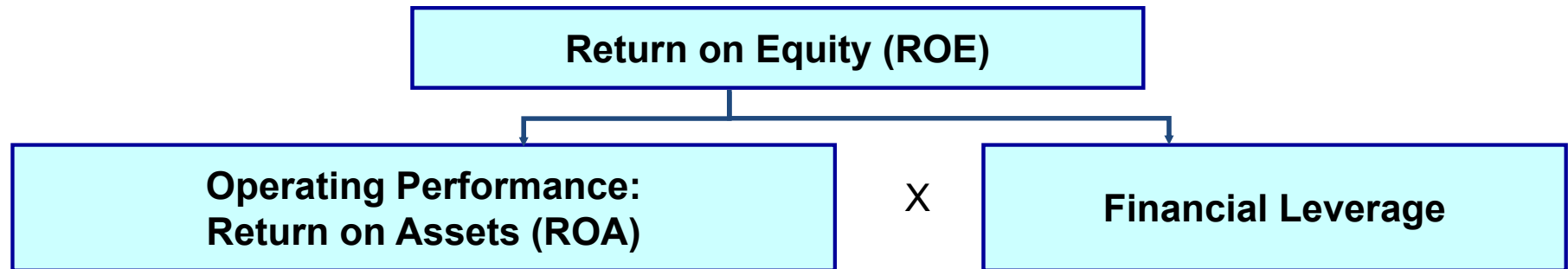
# Drivers of ROE

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- **Two drivers of Return on Equity**
- **Operating performance**
  - How effectively do managers use company resources (assets) to generate profits?
  - Return on Assets (ROA)
  - $ROA = \text{Net Income} / \text{Average Assets}$
- **Financial leverage**
  - How much do the managers use debt to increase available assets for a given level of shareholder investment?
  - Financial leverage =  $\text{Avg. Assets} / \text{Avg. Shareholders' Equity}$
  - Note that this is different from many “leverage” ratios you hear about (e.g., debt-to-equity)

## ROE Framework

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**ROE = Net Income/Assets**

**x Assets/Equity**

**Example: Company raises \$100 from shareholders and borrows \$100 from bank to buy \$200 of assets, which are used to generate \$10 of net income**

**ROE = 10% (10/100), ROA = 5% (10/200), Leverage = 2 (200/100)  
=> 10% = 5% x 2**

# Return on Assets

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- **Two drivers of Return on Assets**
- **Profitability**
  - How much profit does the company earn on each dollar of sales?
  - Return on Sales (ROS)
  - $ROS = \text{Net Income} / \text{Sales}$
- **Efficiency**
  - How much sales does the company generate based on its available resources?
  - Asset Turnover
  - $ATO = \text{Sales} / \text{Avg. Assets}$

## ROA and Leverage

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- Ideally, ROA would measure operating performance independent of the company's financing decisions
- But, the numerator of ROA, Net Income, includes Interest Expense
  - More leverage => higher Interest Expense => lower Net Income
- To truly remove all financing effects from ROA, we must de-lever Net Income
- $ROA = \text{De-Levered Net Income} / \text{Avg. Assets}$
- $\text{De-levered Net Income} = \text{Net Income} + (1-t) \times \text{Interest Expense}$

## ROA Example

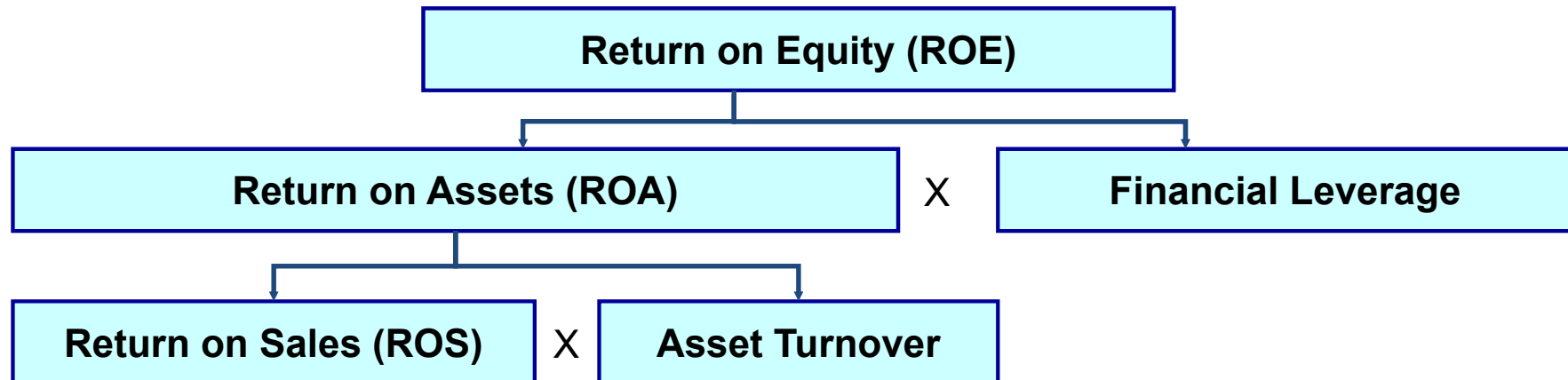
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De-levering NI removes effects of capital structure:

|                             | <u>No debt</u> | <u>Some debt</u>            |
|-----------------------------|----------------|-----------------------------|
| Pretax, pre-interest income | 300            | 300                         |
| Interest expense            | <u>0</u>       | <u>(50)</u>                 |
| Pretax income               | 300            | 250                         |
| Taxes (35%)                 | <u>(105)</u>   | <u>(87.5)</u>               |
| Net income                  | 195            | 162.5                       |
| De-levered Net Income       | 195            | 195 $[162.5 + 50(1 - .35)]$ |

# DuPont Ratio Analysis Framework

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$$\begin{aligned} \text{ROE} &= \text{Net Income/Sales} \times \text{Sales/Assets} \times \text{Assets/Equity} \\ &= \text{Profitability} \times \text{Efficiency} \times \text{Leverage} \end{aligned}$$